

Catapult_Sports Science Document_OpenField

Topic	Description	Internal Guides	Research Papers
Catapult Specific Feature			
Player Load	• Volume parameter; Intensity: Player Load per minute • Highly individual • Parameter which tells you how much work the player did during a training session or game. • Arbitrary unit, meaning it is to be used as a scaling factor to compare the total volume of work performed • Advantages over using distance as an effort metric because it accumulates during tackles, ruck work, or other non-running activities • Based across all axes of the internal tri-axial accelerometer (100 Hz)	• Player Load_Video • Player Load_What is it?	• Barrett, S. et. al (2014). PlayerLoad Reliability, Convergent Validity and Influence of Unit Position During Treadmill Running. <i>Int J Sports Physiol Perform</i>, 9(6):945-52. • Barrett, S. et al. (2016): The within-match patterns of locomotor efficiency during Professional Soccer match play Implications for Injury risk. <i>J Sci Med Sport</i>, 19(10): 810-5.
Football Movement Profile (FMP)	• Based on Player Load • Mechanical work done by athletes • Highly individual, baseline level crucial • Classification: - Very Low Duration & Low Duration - Running Medium Duration & Running High Duration (Linear) - Dynamic Medium Duration & Dynamic High Duration (Multidirectional)	• FMP_Introduction • FMP_WHY • FMP_How to detect? • FMP_Webinar	
Repeat High Intensity Efforts (RHIEs)	• Ability to identify periods where athletes have put in repeated high intensity velocity efforts without adequate recovery • Provides feedback on the relative fitness of an athlete	• Repeat High Intensity Efforts	• Couderc, A. et al. (2019). Repeated High-Intensity Effort Activity in International Male Rugby Sevens. <i>J Strength Cond Res</i>.
Maximum Intensity Intervals (MII)	• Time periods of peak physical demands that are identified based on distance and player load • <i>Worst Case Scenario</i> or <i>Peak Match Demands</i> • Set benchmarks or targets in training	• Maximum Intensity Intervals	• Novak, A. et al. (2021). Analysis of the worst-case scenarios in an elite football team: Towards a better understanding and application. <i>Journal of Sports Sciences</i>. • Martín-García, A. et al. (2020). Physical Demands of Ball Possession Games in Relation to the Most Demanding Passages of a Competitive Match. <i>Journal of sports science & medicine</i>, 19(1), 1–9.

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Metabolic Power	- Estimates the energy demands of acceleration and deceleration events derived from GPS and is differentiated from the Velocity data before any filters are applied to smooth the data - As athletes may operate at an intensity that is greater than the athletes' maximal oxygen consumption, thus making it more practical for measuring Energy costs than measuring the oxygen consumption in high intensity sports	What is Metabolic Power?	Osgnach, C. et al. (2009). Energy Cost and Metabolic Power in Elite Soccer: A New Match Analysis Approach. <i>Medicine & Science in Sports & Exercise</i>. pp 170-178.
Running Symmetry	- Measure of load imbalance between the right and left leg when running - Baseline level crucial - Very useful for rehab - For optimal data: linear runs, constant speed, device fit, surface	Running Symmetry	Saba, D. (2016). Validation of Running Symmetry Using Trunk Mounted Accelerometry: Clinical Trial and Case Study.
Heart Rate Exertion (HRE)	- Comparable with training impulse, parameter to assess cardiovascular load - Internal load that can determine the overall outcome of an athlete's subsequent adaptation to stress - Multiplies time spent in consecutive HR bands by a weighting factor, where time in higher HR bands will be weighted more heavily - Heart Rate bands can be absolute or (more common) relative, meaning calculated as a percentage of the individual's maximum heart rate	Heart Rate Exertion (Part1) Heart Rate Exertion (Part 2)	Lacome, M. et al. (2018). Monitoring Players' Readiness Using Predicted Heart-Rate Responses to Soccer Drills. <i>Int J Sports Physiol Perform</i>, 13(10):1273-1280.
Multi-mechanical models	- Developing a percentage score from volume and intensity in comparison to match demands - Equation via customized parameters	Multi-mechanical models Webinar	Owen, A. et al. (2017). A contemporary multi-modal mechanical approach to training monitoring in elite professional soccer. <i>Science and Medicine in Football</i>. 1-6.

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	DIVE DIRECTION TIME TO FEET DIVE LOAD PRE DIVE LOAD + DIVE IMPACT + POST DIVE LOAD			
Dive Direction	Improved classification of dives with the addition of central dive count Analysed the time it takes following a dive to return to an upright position Better understand the load placed on goalkeepers during dives taking into account: pre impact and post dive activities The sum of all movements of the goalkeepers during the second before impact + How hard the goalkeeper hits the floor + The sum of all movements of the goalkeeper during the second after impact			
Time to Feet	Hills, S. et al. (2020). Profiling the Post-match Top-up Conditioning Practices of Professional Soccer Substitutes: An Analysis of Contextual Influences. <i>J Strength Cond Res</i>, 34:10, pp 2805-2814.			
Dive Load	White, A. et al. (2020). The physical demands of professional soccer goalkeepers throughout a week-long competitive microcycle and transiently throughout match-play. <i>J Sports Sci</i>. 38(8):848-854.			

Topic	Description	Internal Guides	Research Papers
General			
Which metrics to use?	• Volume vs. Intensity • Locomotive vs. Mechanical vs. External	• Which metrics? (Part 1) • Which metrics? (Part 2) • OF Parameters	• Ravé, G. et al. (2020). How to Use Global Positioning Systems (GPS) Data to Monitor Training Load in the “Real World” of Elite Soccer. <i>Front. Physiol.</i> 11:944.
GPS/GNSS	• Athletes change position over time, their movement profiles (and thus displacement/distance covered) can be plotted by regular repetition of this process. As timestamps are simultaneously transmitted, distance covered can be represented relative to time as speed profiles • Measuring on 10Hz	• Wearable Tracking Technology • Wearable Technology Science for Sport	• Malone et al. (2018). Seasonal Training Load and Wellness Monitoring in a Professional Soccer Goalkeeper. <i>Int J Sports Physiol Perform.</i> 13(5):672-675.
Inertial Movement Analysis (IMA)	• Based on data collected from both a tri-axial accelerometer and a tri-axial gyroscope to graphically evaluate the magnitude of every micro movement an athlete makes • Based on 100Hz data and without GPS	• Why IMA? • What is IMA?	• Roell, M. et al. (2018). Player Monitoring in Indoor Team Sports: Concurrent Validity of Inertial Measurement Units to Quantify Average and Peak Acceleration Values. <i>Front. Physiol.</i> 9:141.
Accelerometer	• Measuring accelerations and force at 100 Hz • Showing the exact movement of the unit in all three planes: x plane: left & right movements, y plane: up and down movements, z plane: movements back and forward	• How do accelerometer, gyroscope and magnetometer work?	
Gyroscope	• Measuring speed of rotation at 100 Hz • Tri-axial (yaw, pitch and roll)		
Magnetometer	• Measuring body orientation in three axes at 100 Hz • Electronic compass that help to understand orientation in relation to magnetic north, provide information regarding direction and orientation, helping practitioners to understand the volume of key movements such as changes of direction		
Bands	• The amount of distance or time an athlete spends within user-defined speed brackets		
Dwell Time	• The minimum time a player has to spend in a specific band to count an effort • Example: if Dwell Time is 1 sec, a player needs to be at least 1 second above 20 km/h to get an effort in that zone		
Threshold	• Separation between different zones		